

GitHub Actions & Workflow Generation

DevOps & Infrastructure | Intermediate

Agenda

Time	Topic
0:00–0:08	Actions model
0:08–0:18	Generate CI
0:18–0:28	Debug a failed step
0:28–0:38	Reusable and custom actions
0:38–0:50	Bounded pipeline demo
0:50–1:00	Security, <code>gh-aw</code> , lab handoff

Actions model

Part	Meaning
Workflow	YAML automation in <code>.github/workflows/</code>
Event	Trigger such as <code>push</code> or <code>pull_request</code>
Job	Steps on a runner
Step	An action or command
Secret	Sensitive value outside source
Artifact	Files retained from a job

Generate from repository facts

Read package.json and existing tests. Create ci.yml for pull requests and main.
Use Node.js 20, npm ci, existing lint and test scripts, npm caching, contents: read,
and a test-results artifact. Do not add dependencies or deployment steps.

Check every assumption against the project.

Debug with useful context

Provide the failed step, error, and workflow file. Ask for the smallest fix and the follow-up check.

This workflow fails at "Run tests." Here is the step output and ci.yml.
Identify the cause, propose the smallest fix, and explain what to verify.

Security review

- Set explicit least-privilege `permissions`.
- Keep values in secrets or use OIDC.
- Review triggers, especially `pull_request_target`.
- Pin approved third-party actions to immutable SHAs in production.
- Check commands, runners, artifacts, concurrency, and deployment gates.

```
permissions:  
  contents: read
```

Agentic Workflows

```
gh extension install github/gh-aw
gh aw init daily-repo-status
gh aw compile daily-repo-status
gh aw logs daily-repo-status
```

Review Markdown source and compiled `.lock.yml`. Keep permissions read-only unless a reviewed `safe-outputs` write is required.

Lab

Build CI, design CD, fix supplied broken workflows, and create a JavaScript action. `gh-aw` is optional and requires approval.